

João Monteiro

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RESEARCH INTERESTS

Efficient language model inference (KV-cache compression, discrete diffusion, multi-token prediction), search and retrieval.

EXPERIENCE

Apple MLR, August 2025 - Present

Senior Research Scientist
Paris, France

Role details:

- Fundamental research with a focus on efficient autoregressive inference (KV-cache compression and reuse), (multimodal) discrete diffusion, and search/retrieval.

Autodesk AI Lab,

September 2024 - August 2025

Principal AI Research Scientist
London, UK

Role details:

- Applied research on NLP. Train language models on 3d geometry data.

ServiceNow Research,

December 2021 - September 2024

Staff Research Scientist
London, UK

Role details:

- Applied research on NLP. Train large-scale models on distributed settings.
- Co-authored BigCode's STARCORDER (780K+ downloads: <https://huggingface.co/bigcode/starcoder>) and led the training of STARENCORDER, a code encoder used for code search and PII detection and removal (294K+ downloads: <https://huggingface.co/bigcode/starencoder>).
- Fundamental research with a focus on OOD and adversarial robustness.
- Advise students/interns and establish academic collaborations.
- Design and conduct technical interviews.

Borealis AI,

May 2021 - October 2021

Research intern
Montreal, Canada

Role details:

- Conducted research on monotonic model classes for tabular data.
- Helped deploy and open sourcing the approaches we developed.

Google,

September 2020 - April 2021

Student researcher
Montreal, Canada

Role details:

- Led a research effort on out-of-distribution robustness.
- Introduced domain-conditional predictors, able to adapt to varying properties of input data sources.

SELECTED PUBLICATIONS

- **J. Monteiro**, L. Béthune, A. Filippova, S. Laguna, D. Grangier, M. Cuturi “KV-Kaizen: Learning Context-Adaptive Cache Compression Choices”, *Pre-print*.
- M. Madeira, A. Shidani, A. Bizeul, V. Turrise, L. Béthune, B. Devnani, D. Busbridge, P. Ablin, **J. Monteiro** “On Trajectory-Aware Training for Masked Diffusion Language Models”, *Pre-print*.
- S. Laguna, **J. Monteiro**, M. Cuturi, P. Ablin, E. Gualdoni “Cartridges++: KV Cache Compression without Off-Context Derailment”, *Pre-print*.
- V. Castin, K. Sakamoto, A. Filippova, **J. Monteiro**, M. Cuturi, P. Ablin “KV-Lingo: Learning KV-Cache Translators with Distillation”, *Pre-print*.
- **J. Monteiro**, M. Klein, P. Ablin, M. Cuturi “Nectar: Neural Estimation of Cached-Token Attention via Regression”, *Pre-print*.
- “The Design Space of Tri-Modal Masked Diffusion Models”, **NeurIPS 2026, Spotlight**. **Lengthy author list omitted*.
- T. X. Olausson, **J. Monteiro**, M. Klein, M. Cuturi “Amortizing Maximum Inner Product Search with Learned Support Functions”, **ICML 2026**.
- M. Jazbec, T. X. Olausson, L. Béthune, P. Ablin, M. Kirchhof, **J. Monteiro**, V. G. T. da Costa, J. Ramapuram, M. Cuturi “Learning Unmasking Policies for Diffusion Language Models”, **ICML 2026, Oral**.
- A. Filippova, D. Grangier, M. Cuturi, **J. Monteiro** “Stochastic KV Routing: Enabling Adaptive Depth-Wise Cache Sharing”, *Pre-print*.
- C. Tsigotis, V. Adlakha, **J. Monteiro**, A. Courville, P. Taslakian “BiXSE: Improving Dense Retrieval via Probabilistic Graded Relevance Distillation”, **COLM 2025**.
- “BigDocs: An Open and Permissively-Licensed Dataset for Training Multi-modal Models on Document and Code Tasks”, **ICLR 2025**. **Lengthy author list omitted*.
- **J. Monteiro**, P. A. Noël, E. Marcotte, S. Rajeswar, V. Zantedeschi, D. Vázquez, N. Chapados, C. Pal, P. Taslakian “RepLiQA: A Question-Answering Dataset for Benchmarking LLMs on Unseen Reference Content”, **NeurIPS 2024**.
- C. Guille-Escuret, P. A. Noël, D. Vázquez, I. Mitliagkas, **J. Monteiro** “Expecting The Unexpected: Towards Broad Out-Of-Distribution Detection”, **NeurIPS 2024**.
- **J. Monteiro**, E. Marcotte, P. A. Noël, V. Zantedeschi, D. Vázquez, N. Chapados, C. Pal, P. Taslakian “XC-Cache: Cross-Attending to Cached Context for Efficient LLM Inference”, **EMNLP (Findings) 2024**.
- C. Guille-Escuret, P. Rodríguez, D. Vázquez, I. Mitliagkas, **J. Monteiro** “CADet: Fully Self-Supervised Out-Of-Distribution Detection With Contrastive Learning”, **NeurIPS 2023**.
- **J. Monteiro**, P. Rodríguez, P. A. Noël, I. Laradji, D. Vázquez “Constraining Representations Yields Models That Know What They Don’t Know”, **ICLR 2023**.
- “StarCoder: may the source be with you!”, Transactions on Machine Learning Research, 2023. **TMLR**. **Lengthy author list omitted*.
- **J. Monteiro**, I. Albuquerque, J. Alam, R. D. Hjelm, T. Falk “An end-to-end approach for the verification problem: learning the right distance”, **ICML 2020**.
- **J. Monteiro***, I. Albuquerque*, T. Doan, B. Considine, T. Falk, I. Mitliagkas, “Multi-objective training of Generative Adversarial Networks with multiple discriminators”, **ICML 2019**. **Equal contribution*

EDUCATION	University of Quebec - INRS Ph.D. Montreal, Canada	January 2017 - November 2021
	University of Pernambuco Master's degree, Computer Engineering Recife, Brazil	August 2015 - December 2016
	University of Pernambuco Mechanical Engineering Recife, Brazil	August 2007 - June 2012

ADVISED STUDENTS

- Supervised Manuel Madeira, research intern at Apple MLR. Manuel led our work on trajectory-aware training of masked diffusion language models.
- Hosted Aarash Feizi, visiting researcher at ServiceNow based in Montreal. Aarash is working on what we call large language kernels, where we prompt LLMs to measure the similarity between data points. In particular, we are using this setting to test for the ability of LLMs to behave as automatic evaluators, and determine how different prompting strategies yield different auto-evaluation behaviours, *e.g.*, a very harsh vs. a more nuanced/smooth evaluator.
- Hosted Christos Tsirigotis, visiting researcher at ServiceNow based in Montreal. Christos worked on *post-hoc* logit adjustment approaches to yield classifiers able to handle systematic distribution shifts. Currently, we are exploring information retrieval approaches that rely less on re-ranking strategies.
- Hosted Xiangru Jian, visiting researcher at ServiceNow based in Waterloo. Xiangru has been focusing on data curation. Our goal is to put together large-scale high quality multi-modal datasets that enable different capabilities on models trained on them.
- Hosted Mehrnaz Mofakhami, visiting researcher at ServiceNow based in Montreal. With Mehrnaz, we proposed early-exit approaches to improve the inference efficiency of large models. Crucially, we showed that scaling up model size can reduce inference cost under early-exiting.
- Hosted Charles Guille-Escuret, visiting researcher at ServiceNow based in Montreal. Charles worked on self-supervised learning of data-dependent kernels for 2-sample tests and out-of-distribution detection.

ENGINEERING

- Extensive experience with PyTorch. Recently moved to JAX.
- Trained billion-parameter models in trillion-token datasets in distributed settings.

OTHER

- *Academic service*: Area chair for ICLR 2024-2027, ICML 2025-2026, and NeurIPS 2026. Regularly reviewing for flagship Machine Learning conferences since 2020. Mentor for the WiML workshop at NeurIPS 2024 on Interpretability and Explainability.
- *Teaching*: 3-day course on language models training and evaluation at the Bertinoro Spring School 2025, organized by the University of Bologna. <https://tempesta.cs.unibo.it/projects/BISS/2025/>
- *Invited talks*: An intro to language models and challenges in applying them to AEC domain specific languages (Invited talk at *2300: Technology and Practice at Yale Architecture* - February 2025). RepLiQA: a robust benchmark for QA (<https://www.youtube.com/watch?v=qgzAD6pjvq4>). Closing the gap between machine learning research and practice via versatile and robust predictors (ServiceNow - July 2021).